

Vector Spaces (of Functions)

Idea: build complex functions from simple ones.

Taylor

$$f(x) \approx f(x_0) \mathbf{1} + f'(x_0)(x-x_0) + \dots + \frac{1}{n!} f^{(n)}(x_0)(x-x_0)^n$$

Fourier

$$f(x) \approx \hat{f}_0 \mathbf{1} + \hat{f}_1 e^{ix} + \hat{f}_{-1} e^{-ix} + \dots + \hat{f}_n e^{inx} + \hat{f}_{-n} e^{-inx}$$

In both examples, $f(x)$ is built up using linear combinations of simpler functions.

$$(*) \quad f(x) \approx c_1 e_1(x) + \dots + c_n e_n(x)$$

$\hat{=}$ dictionary $\rightarrow$

$\Rightarrow$ How to choose coefficients $c_1, \dots, c_n$?

Idea: use tools from linear algebra to construct and study approximations (*).

Q: How can we apply tools from linear algebra to functions?

Example 1: Interpolation/Regression

Suppose we have samples of $f: [0, 1] \rightarrow \mathbb{R}$.

$$f_1 = f(x_1), f_2 = f(x_2), \dots, f_m = f(x_m)$$

If we want (*) to fit the data well, then

$$f_1 \approx c_1 e_1(x_1) + c_2 e_2(x_1) + \dots + c_n e_n(x_1)$$

$$f_2 \approx c_1 e_1(x_2) + c_2 e_2(x_2) + \dots + c_n e_n(x_2)$$

$\vdots$

$$f_m \approx c_1 e_1(x_m) + c_2 e_2(x_m) + \dots + c_n e_n(x_m)$$

Equivalently, in matrix form

$$\begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_m \end{bmatrix} \approx \begin{bmatrix} e_1(x_1) & \dots & e_n(x_1) \\ e_1(x_2) & \dots & e_n(x_2) \\ \vdots & & \vdots \\ e_1(x_m) & \dots & e_n(x_m) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

$\underline{f} \quad \approx \quad E \quad \underline{c}$

$\Rightarrow$ "Solve" linear system for coefficients $\underline{c}$.

Vector Spaces

A vector space is a nonempty set V , closed under addition and scalar multiplication.

Vector Addition

$$x, y \in V \Rightarrow x + y \in V$$

$$(x + y) + z = x + (y + z)$$

$$x + y = y + x$$

$$\exists 0 \in V \text{ s.t. } x + 0 = x \quad \forall x \in V$$

$$\forall x \in V \exists -x \text{ s.t. } x + (-x) = 0$$

Scalar Multiplication

$$x \in V \Rightarrow \bigwedge_{\alpha \in \mathbb{R} \text{ or } \mathbb{C}} \alpha x \in V$$

$$\alpha(\beta x) = (\alpha\beta)x$$

$$\alpha(x + y) = \alpha x + \alpha y$$

$$(\alpha + \beta)x = \alpha x + \beta x$$

$$1x = x$$

Essentially,

$$\underbrace{\alpha x + \beta y + \dots + \gamma z}_{\text{finite linear combo}} \in V$$

finite linear combo
always in the space V

example: $C[-1, 1] = \{f: [-1, 1] \rightarrow \mathbb{R} \mid f \text{ continuous on } [-1, 1]\}$

E

$$h(x) = \alpha f(x) + \beta g(x) \quad \text{also continuous}$$

Question: Can you think of other vector spaces whose elements are functions?

Example: $C^n[-1,1] = \{f \in C[-1,1] \mid f', f'', \dots, f^{(n)} \in C[-1,1]\}$

$$(k=1, 2, \dots, n) \quad \frac{d^k}{dx^k} (\alpha f(x) + \beta g(x)) = \alpha \frac{d^k f}{dx^k}(x) + \beta \frac{d^k g}{dx^k}(x)$$

$\Rightarrow$ If f, g have n continuous derivatives, then all linear combinations do too.

Example: $P_n = \{p \in C[-1, 1] \mid p(x) = a_0 + a_1x + \dots + a_nx^n\}$

$\uparrow \quad \uparrow \quad \uparrow$
scalar coeffs

$$= a_0 + a_1 x + \dots + a_n x^n$$

$$\alpha p(x) + \beta q(x) = (\alpha a_0 + \beta b_0) + \dots + (\alpha a_n + \beta b_n)x^n$$

$\Rightarrow$ Combinations of degree (at most) n poly's are also degree (at most) n poly's.

The last two examples are subspaces of $C[-1, 1]$ = space of continuous functions.

A **subspace** W of V is a nonempty set $W \subseteq V$ that is closed under linear combos.

Questions: What are other subspaces of $C[-1,1]$?

Zero at boundaries $\Rightarrow C_0[-1,1] = \{f \in C[-1,1] \mid f(-1) = f(1) = 0\}$

odd functions $\Rightarrow C_o[-1,1] = \{f \in C[-1,1] \mid f(-x) = -f(x) \quad \forall x \in [-1,1]\}$

Dimension, Span, Dependence

It's useful to have a sense of the intrinsic "size" of a vector space or subspace. In some sense, how much information do we need to store and manipulate its elements?

A set $S = \{x_1, \dots, x_n\} \subset V$ is called **linearly independent** if the only way to get

$$\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n = 0$$

is if all of the scalars $\alpha_1, \alpha_2, \dots, \alpha_n = 0$.

Otherwise, S is called **linearly dependent**.

The span of $S = \{x_1, x_2, \dots, x_n\} \subseteq V$ is the subspace formed via linear combos

$$W = \{x \in V \mid x = \alpha_1 x_1 + \dots + \alpha_n x_n\}$$

all possible linear combos

Question: Is $\{1, x, \dots, x^n\}$ linearly indep.?

Yes! $0 = \alpha_1 \cdot 1 + \alpha_2 \cdot x + \dots + \alpha_n x^n \Rightarrow \alpha_1 = \alpha_2 = \dots = \alpha_n = 0$

Question: What is the span of $\{1, x, \dots, x^n\}$?

$$\begin{aligned} \text{span}\{1, x, \dots, x^n\} &= \{f \in \mathbb{C}[1, 1] \mid a_0 + a_1 x + \dots + a_n x^n\} \\ &= \mathbb{P}_n \end{aligned}$$

A set $S \subseteq V$ is a **basis** for V if

i) linearly independent, and

ii) the span of S is V .

If S is a basis for V , then for each $x \in V$, there is a unique set of scalars s.t.

$$x = \alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n$$

↖ ↑ ↗
 unique coordinates
 of x in basis S

Any basis S for V has the same # of elements and this # is the dimension of V .

If V has no finite basis set, it is infinite-dimensional. In this course, all infinite-dimensional spaces of interest will have a countable basis set $\{e_j\}_{j=1}^{\infty}$, i.e., a basis with elements indexed by integers.

Example: The vector space of polynomials $\mathcal{P} = \bigcup_{n=0}^{\infty} \mathcal{P}_n$ is infinite-dimensional with basis

$$S = \{x^k\}_{k=0}^{\infty}.$$